from McGreevy's lectures on OFT

2.2.8 Coherent state path integrals for fermions

[Shankar, *Principles of QM*, path integrals revisited. In this chapter of his great QM textbook, Shankar sneaks in lots of insights useful for modern condensed matter physics]

Consider a single fermion mode operator (as in our discussion of Jordan-Wigner in 0+1 dimensions);

$$\{\mathbf{c},\mathbf{c}\}=0,\quad \{\mathbf{c}^\dagger,\mathbf{c}^\dagger\}=0,\quad \{\mathbf{c},\mathbf{c}^\dagger\}=1\ .$$

With a single mode, the general Hamiltonian is

$$\mathbf{H}=\mathbf{c}^\dagger\mathbf{c}\,(\omega_0-\mu)$$

(ω_0 and μ are (redundant when there is only one mode) constants). We might be interested in its thermal partition function

$$Z=\mathrm{tr}\ e^{-\frac{\mathbf{H}}{T}}\ .$$

(In this example, it happens to equal $Z=1+e^{-\frac{\omega_0-\mu}{T}}$, as you can see by computing the trace in the eigenbasis of $\mathbf{n}=\mathbf{c}^\dagger\mathbf{c}$. But never mind that; the one mode is a proxy for many, where it's not quite so easy to sum.) How do we trotterize this? That is, what is the corresponding classical system? One answer is to use the 0d Jordan-Wigner map from §2.2.5. That route takes us back where we came from. Here's another answer. We can do the Trotterizing using any resolution of the identity on $\mathcal{H}$, so there can be many very-different-looking answers to this question.

Let's define coherent states for fermionic operators:

$$\mathbf{c}|\psi\rangle=\psi|\psi\rangle. \tag{49}$$

Here ψ is a c-number (not an operator), but acting twice with $\mathbf{c}$ we see that we must have $\psi^2=0$. ψ is a grassmann number. These satisfy

$$\psi_1\psi_2=-\psi_2\psi_1,\psi\mathbf{c}=-\mathbf{c}\psi \tag{50}$$

– they anticommute with each other and with fermionic operators, and commute with ordinary numbers and bosons. They seem weird but they are easy. We'll need to consider multiple grassmann numbers when we have more than one fermion mode, where $\{\mathbf{c}_1,\mathbf{c}_2\}=0$ will require that they anticommute $\{\psi_1,\psi_2\}=0$ (as in the definition (50)); note that we will be simultaneously diagonalizing operators which *anti*commute.

It's very easy to solve the equation (49):

$$|\psi\rangle=|0\rangle-\psi|1\rangle$$

where as above $|0\rangle$ is the empty state ($\mathbf{c}|0\rangle=0$) and $|1\rangle=\mathbf{c}^\dagger|0\rangle$ is the filled state. (Check: $\mathbf{c}|\psi\rangle=\mathbf{c}|0\rangle-\mathbf{c}\psi|1\rangle=+\psi\mathbf{c}|1\rangle=\psi|0\rangle=\psi|\psi\rangle$.)

Similarly, the left-eigenvector of the creation operator is

$$\langle\bar{\psi}|\mathbf{c}^\dagger = \langle\bar{\psi}|\bar{\psi}, \quad \langle\bar{\psi}| = \langle 0| - \langle 1|\bar{\psi} = \langle 0| + \bar{\psi}\langle 1|.$$

Notice that these states are weird in that they are elements of an enlarged hilbert space with grassmann coefficients (usually we just allow complex numbers). Also, $\bar{\psi}$ is *not* the complex conjugate of ψ and $\langle\bar{\psi}|$ is not the adjoint of $|\psi\rangle$. Rather, their overlap is

$$\langle\bar{\psi}|\psi\rangle = 1 + \bar{\psi}\psi = e^{\bar{\psi}\psi}.$$

Grassmann calculus summary. In the last expression we have seen an example of the amazing simplicity of Taylor's theorem for grassmann functions:

$$f(\psi) = f_0 + f_1\psi \ .$$

Integration is just as easy and its the same as taking derivatives:

$$\int \psi d\psi = 1, \qquad \int 1 d\psi = 0.$$

With more than one grassmann we have to worry about the order:

$$1 = \int \bar{\psi}\psi d\psi d\bar{\psi} = -\int \bar{\psi}\psi d\bar{\psi} d\psi.$$

The only integral, really, is the gaussian integral:

$$\int e^{-a\bar{\psi}\psi} d\bar{\psi} d\psi = a.$$

Many of these give

$$\int e^{-\bar{\psi}\cdot A\cdot\psi} d\bar{\psi} d\psi = \det A.$$

Here $\bar{\psi}\cdot A\cdot\psi \equiv \left(\bar{\psi}_1, \cdots, \bar{\psi}_M\right) \begin{pmatrix} A_{11} & A_{12} & \cdots \\ A_{21} & \ddots & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_M \end{pmatrix}$. One way to get this expression is to change variables to diagonalize the matrix A.

$$\langle\bar{\psi}\psi\rangle \equiv \frac{\int \bar{\psi}\psi e^{-a\bar{\psi}\psi} d\bar{\psi} d\psi}{\int e^{-a\bar{\psi}\psi} d\bar{\psi} d\psi} = -\frac{1}{a} = -\langle\psi\bar{\psi}\rangle.$$

If for many grassman variables we use the action $S = \sum_i a_i\bar{\psi}_i\psi_i$ (diagonalize A above) then

$$\langle\bar{\psi}_i\psi_j\rangle = \frac{\delta_{ij}}{a_i} \equiv \langle\bar{i}j\rangle \tag{51}$$

and Wick's theorem here is

$$\langle\bar{\psi}_i\bar{\psi}_j\psi_k\psi_l\rangle = \langle\bar{i}l\rangle\langle\bar{j}k\rangle - \langle\bar{i}k\rangle\langle\bar{j}l\rangle.$$

Back to quantum mechanics: The resolution of $\mathbb{1}$ in this basis is

$$\mathbb{1} = \int d\bar{\psi} d\psi \; e^{-\bar{\psi}\psi} |\psi\rangle\langle\bar{\psi}| \tag{52}$$

And if $\mathbf{A}$ is a bosonic operator (made of an even number of grassmann operators),

$$\mathrm{tr}\mathbf{A} = \int d\bar{\psi} d\psi \; e^{-\bar{\psi}\psi} \langle -\bar{\psi}|\mathbf{A}|\psi\rangle \; .$$

(Note the minus sign; it will lead to a deep statement.) So the partition function is:

$$Z = \int d\bar{\psi}_0 d\psi_0 \; e^{-\bar{\psi}_0\psi_0} \langle -\bar{\psi}_0| \underbrace{e^{-\frac{\mathbf{H}}{T}}}_{=\underbrace{(1-\Delta\tau\mathbf{H})\cdots(1-\Delta\tau\mathbf{H})}_{M \text{ times}}} |\psi_0\rangle$$

Now insert (52) in between each pair of Trotter factors to get

$$Z = \int \prod_{l=0}^{M-1} d\bar{\psi}_l d\psi_l e^{-\bar{\psi}_l\psi_l} \langle\bar{\psi}_{l+1}|(1-\Delta\tau\mathbf{H})|\psi_l\rangle \; .$$

Because of the $-\bar{\psi}$ in (52), to get this nice expression we had to define an extra letter

$$\bar{\psi}_M = -\bar{\psi}_0, \quad \psi_M = -\psi_0 \tag{53}$$

so we could replace $\langle -\bar{\psi}_0| = \langle\bar{\psi}_N|$. [End of Lecture 10]

Now we use the coherent state property to turn the matrix elements into grassmann-valued functions:

$$\langle\bar{\psi}_{l+1}| \left(1 - \Delta\tau H(\mathbf{c}^\dagger, \mathbf{c})\right) |\psi_l\rangle = \langle\bar{\psi}_{l+1}| \left(1 - \Delta\tau H(\bar{\psi}_{l+1}, \psi_l)\right) |\psi_l\rangle \overset{\Delta\tau\to 0}{=} e^{\bar{\psi}_{l+1}\psi_l} e^{-\Delta\tau H(\bar{\psi}_{l+1},\psi_l)}.$$

It was important that in $\mathbf{H}$ all $\mathbf{c}$s were to the right of all $\mathbf{c}^\dagger$s, *i.e.* that $\mathbf{H}$ was normal ordered.)

So we have

$$\begin{aligned} Z &= \int \prod_{l=0}^{M-1} d\bar{\psi}_l d\psi_l e^{-\bar{\psi}_l\psi_l} e^{\bar{\psi}_{l+1}\psi_l} e^{-\Delta\tau H(\bar{\psi}_{l+1},\psi_l)} \\ &= \int \prod_{l=0}^{M-1} d\bar{\psi}_l d\psi_l \exp\left(\Delta\tau \left(\underbrace{\frac{\bar{\psi}_{l+1} - \bar{\psi}_l}{\Delta\tau}}_{=\partial_\tau\bar{\psi}} \psi_l - H(\bar{\psi}_{l+1}, \psi_l)\right)\right) \\ &\simeq \int [D\bar{\psi}D\psi] \exp\left(\int_0^{1/T} d\tau \; \bar{\psi}(\tau)\left(-\partial_\tau - \omega_0 + \mu\right)\psi(\tau)\right) = \int [D\bar{\psi}D\psi] e^{-S[\bar{\psi},\psi]}. \end{aligned} \tag{54}$$

Points to note:

- In the penultimate step we defined, as usual, continuum fields

$$\psi(\tau_l = \Delta\tau l) \equiv \psi_l, \qquad \bar{\psi}(\tau_l = \Delta\tau l) \equiv \bar{\psi}_l.$$

- We elided the difference $H(\bar{\psi}_{l+1}, \psi_l) = H(\bar{\psi}_l, \psi_l) + \mathcal{O}(\Delta\tau)$ in the last expression. This difference is usually negligible and sometimes helpful (an example where it's helpful is the discussion of the number density below).

- The APBCs (53) on $\psi(\tau + \frac{1}{T}) = -\psi(\tau)$ mean that in its fourier representation[22]

$$\psi(\tau) = T\sum_n \psi(\omega)e^{-\mathbf{i}\omega_n\tau}, \qquad \bar{\psi}(\tau) = T\sum_n \bar{\psi}(\omega)e^{\mathbf{i}\omega_n\tau} \tag{55}$$

  the *Matsubara frequencies*

$$\omega_n = (2n+1)\pi T, \quad n \in \mathbb{Z}$$

  are half-integer multiples of πT.

- The measure $[D\bar{\psi}D\psi]$ is defined by this equation, just as in the bosonic path integral.

- The derivative of a grassmann function is also defined by this equation; note that $\psi_{l+1} - \psi_l$ is not 'small' in any sense.

- In the last step we integrated by parts, *i.e.* relabeled terms in the sum, so

$$\sum_l \left(\bar{\psi}_{l+1} - \bar{\psi}_l\right)\psi_l = \sum_l \bar{\psi}_{l+1}\psi_l - \sum_l \bar{\psi}_l\psi_l = \sum_{l'=l-1} \bar{\psi}_{l'}\psi_{l-1} - \sum_l \bar{\psi}_l\psi_l = -\sum_l \bar{\psi}_l\left(\psi_l - \psi_{l-1}\right).$$

  Note that no grassmanns were moved through each other in this process.

The punchline of this discussion for now (for purposes of our study of the scaling limit of the TFIM) is that the euclidean action is

$$\boxed{S[\bar{\psi}, \psi] = \int d\tau \left(\bar{\psi}\partial_\tau\psi + H(\bar{\psi}, \psi)\right)}\,.$$

The first-order kinetic term we've found $\bar{\psi}\partial_\tau\psi$ is sometimes called a 'Berry phase term'. Note the funny-looking sign.

Continuum limit warning (about the red $\simeq$ in (54)). The Berry phase term is *actually*

$$\sum_{l=0}^{N-1} \bar{\psi}_{l+1}\left(\psi_{l+1} - \psi_l\right) = T\sum_{\omega_n} \bar{\psi}(\omega_n)\left(1 - e^{\mathbf{i}\omega_n\tau}\right)\psi(\omega_n)$$

[22] $\bar{\psi}$ is still not the complex conjugate of ψ but the relative sign is convenient.

and in (54) we have kept only the leading nonzero term:

$$\left(1 - e^{\mathbf{i}\omega_n \tau}\right) \to \mathbf{i}\omega_n \tau.$$

Clearly this replacement is just fine if

$$\omega_n \tau \ll 1$$

for all ω_n which matter. Which ω_n contribute? I claim that if we use a reasonable $\mathbf{H} = \mathbf{H}_{\text{quadratic}} + \mathbf{H}_{\text{int}}$, reasonable quantities like Z, $\langle \mathcal{O}^\dagger \mathcal{O} \rangle$, are dominated by $\omega_n \ll \tau^{-1}$.

There's more we can learn from what we've done here that I don't want to pass up. Let's use this formalism to compute the fermion density at $T = 0$:

$$\langle \mathbf{N} \rangle = \frac{1}{Z} \text{tr} e^{-\mathbf{H}/T} \mathbf{c}^\dagger \mathbf{c}.$$

This is an example where the annoying $\Delta\tau$s in the path integral not only matter, but are extremely friendly to us.

Frequency space, $T \to 0$.

Let's change variables to frequency-space fields, which diagonalize S. The Jacobian is 1 (since fourier transform is unitary):

$$D\bar{\psi}(\tau) D\psi(\tau) = \prod_n d\bar{\psi}(\omega_n) d\psi(\omega_n) \overset{T \to 0}{\to} D\bar{\psi}(\omega) D\psi(\omega).$$

The partition function is

$$Z = \int D\bar{\psi}(\omega) D\psi(\omega) \ \exp\left(T \sum_{\omega_n} \bar{\psi}(\omega_n) \left(\mathbf{i}\omega_n - \omega_0 + \mu\right) \psi(\omega_n) \right).$$

Notice that in the zero-temperature limit

$$T \sum_{\omega_n} \mapsto \int \frac{d\omega}{2\pi} \equiv \int đ\omega.$$

(This is the same fact as $V \sum_k \mapsto \int đ^d k$ in the thermodynamic limit.) So the zero-temperature partition function is

$$Z \overset{T \to 0}{=} \int D\bar{\psi}(\omega) D\psi(\omega) \ \exp\left(\int_{-\infty}^{\infty} đ\omega \bar{\psi}(\omega) \left(\mathbf{i}\omega - \omega_0 + \mu\right) \psi(\omega) \right).$$

Using the gaussian-integral formula (51) you can see that the *propagator* for ψ is

$$\langle \bar{\psi}(\omega_1) \psi(\omega_2) \rangle = \underbrace{\frac{\delta_{\omega_1, \omega_2}}{T}}_{\overset{T \to 0}{\to} \delta(\omega_1 - \omega_2)} \frac{2\pi}{\mathbf{i}\omega_1 - \omega_0 + \mu}. \tag{56}$$

In particular $\langle\bar\psi(\omega)\psi(\omega)\rangle = \frac{2\pi/T}{\mathbf{i}\omega-\omega_0+\mu}$. $\delta(\omega=0) = 1/T$ is the 'volume' of the time direction.

Back to the number density. Using the same strategy as above, we have

$$\langle\mathbf{N}\rangle = \frac{1}{Z}\int \prod_{l=0}^{M-1+1}\left(d\bar\psi_l d\psi_l e^{-\bar\psi_l\psi_l}\right)\prod_{l=1}^{M-1}\langle\bar\psi_{l+1}|(1-\Delta\tau\mathbf{H}(\mathbf{c}^\dagger\mathbf{c}))|\psi_l\rangle \quad \underbrace{\langle\bar\psi_{N+1}|\mathbf{c}^\dagger\mathbf{c}|\psi_N\rangle}_{=\bar\psi_{N+1}\psi_N=\bar\psi(\tau_N+\Delta\tau)\psi(\tau_N)} \quad .$$

This formula has a built-in point-splitting of the operators!

$$\begin{aligned}\langle\mathbf{N}\rangle &= \frac{1}{Z}\int D\bar\psi D\psi\; e^{-S[\bar\psi,\psi]}\bar\psi(\tau_N+\Delta\tau)\psi(\tau_N)\\ &= \int_{-\infty}^{\infty}đ\omega\frac{e^{\mathbf{i}\omega\Delta\tau}}{\mathbf{i}\omega-\omega_0+\mu} = \theta(\mu-\omega_0).\end{aligned} \tag{57}$$

Which is the right answer: the mode is occupied in the groundstate only if $\omega_0 < \mu$. In the last step we used the fact that $\Delta\tau > 0$ to close the contour in the UHP; so we only pick up the pole if it is in the UHP. Notice that this quantity is very *UV sensitive*: if we put a frequency cutoff on the integral, $\int^\Lambda \frac{d\omega}{\omega} \sim \log\Lambda$, the integral diverges logarithmically. For most calculations the $\Delta\tau$ can be ignored, but here it told us the right way to treat the divergence. [23]

Bosons. This is a good place to mention quickly that we can do the same thing for bosons, using ordinary SHO (simple harmonic oscillator) coherent states. What I mean by 'bosons' is a many-body system whose Hilbert space can be written as $\mathcal{H} = \otimes_k\mathcal{H}_k$ where k is a label (could be real space, could be momentum space) and $\mathcal{H}_k = \text{span}\{|0\rangle_k, \mathbf{a}_k^\dagger|0\rangle_k, \frac{1}{2}\left(\mathbf{a}_k^\dagger\right)^2|0\rangle_k, ...\}$ is the SHO Hilbert space.

For each normal mode $\mathbf{a}$, coherent states are

$$\mathbf{a}|\phi\rangle = \phi|\phi\rangle; \qquad |\phi\rangle = \mathcal{N}e^{\phi\mathbf{a}^\dagger}|0\rangle.$$

The eigenbra of $\mathbf{a}$ is $\langle\phi|$, with

$$\langle\phi|\mathbf{a} = \langle\phi|\phi^\star, \quad \langle\phi| = \langle 0|e^{+\phi^\star\mathbf{a}}\mathcal{N}.$$

[23] The calculation between the first and second lines of (57) is familiar to field theorists – it can be described as a feynman diagram with one line between the two insertions. More prosaically, it is

$$\langle\bar\psi(\tau_N+\Delta\tau)\psi(\tau_N)\rangle \overset{(55)}{=} T^2\sum_{nm}e^{\mathbf{i}(\omega_n-\omega_m)\tau+\mathbf{i}\omega_n\Delta\tau}\langle\bar\psi(\omega_n)\psi(\omega_m)\rangle \overset{(56)}{=} T\sum_m\frac{e^{\mathbf{i}\omega_n\Delta\tau}}{\mathbf{i}\omega_n-\omega_0+\mu} \overset{T\to 0}{\to} \int đ\omega\frac{e^{\mathbf{i}\omega\Delta\tau}}{\mathbf{i}\omega-\omega_0+\mu}.$$

Their overlap is[24]:

$$\langle\phi_1|\phi_2\rangle = e^{\phi_1^\star\phi_2}.$$

If we choose $\mathcal{N} = e^{-|\phi|^2/2}$, they are normalized, but it is more convenient to set $\mathcal{N} = 1$. The overcompleteness relation on $\mathcal{H}_k$ is

$$\mathbb{1}_k = \int \frac{d\phi d\phi^\star}{\pi} e^{-|\phi|^2} |\phi\rangle\langle\phi|.$$

It will be convenient to arrange all our operators into sums of *normal-ordered* operators:

$$:\mathbf{a}_k\mathbf{a}_l^\dagger := :\mathbf{a}_l^\dagger\mathbf{a}_k := \mathbf{a}_l^\dagger\mathbf{a}_k$$

with all annihilation operators to the right of all creation operators. Coherent state expectation values of such operators can be built from the monomials

$$\langle\phi|\prod_k \left(\mathbf{a}_k^\dagger\right)^{M_k} (\mathbf{a}_k)^{N_k} |\phi\rangle = \prod_k (\phi_k^\star)^{M_k} (\phi_k)^{N_k}.$$

Let the Hamiltonian be $\mathbf{H} = H(\{\mathbf{a}_k^\dagger\},\{\mathbf{a}_k\}) =: \mathbf{H} :$, normal ordered. By now you know how to derive the path integral using this resolution of the identity:

$$\begin{aligned} Z &= \mathrm{tr} e^{-\mathbf{H}/T} \\ &= \int_{\phi_{N+1}=\phi_0} \prod_{l=0}^{N} d\phi_l \; e^{-\sum_{l=0}^{N}\left(\phi_{l+1}(\phi_{l+1}-\phi_l)-\Delta\tau H(\phi_{l+1}^\star\phi_l)\right)} \\ &\simeq \int_{\phi(0)=\phi(1/T)} [D\phi] \; e^{-\int_0^{1/T} d\tau(\phi^\star\partial_\tau\phi + H(\phi^\star,\phi))}. \end{aligned} \tag{58}$$

Real time. If you are interested in real-time propagation, rather than euclidean time, just replace the euclidean propagator $e^{-\tau\mathbf{H}} \mapsto e^{-\mathbf{i}t\mathbf{H}}$. The result, for example, for the amplitude to propagate from one bose coherent state to another is

$$\langle\phi_f, t_f|e^{-\mathbf{i}t\mathbf{H}}|\psi_0, t_0\rangle = \int_{\phi(t_0)=\phi_0}^{\phi(t_f)=\phi_f} D\phi^\star D\phi \; e^{\frac{\mathbf{i}}{\hbar}\int_{t_0}^{t_f} dt(\mathbf{i}\hbar\phi^\star\partial_t\phi - H(\phi,\phi^\star))}.$$

Note the distinguishing feature of the Berry phase term that it produces a complex term in the real-time action.

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[24] Check this by expanding the coherent states in the number basis and doing the integrals

$$\int \frac{d\phi d\phi^\star}{\pi} e^{-\phi\phi^\star} \phi^n (\phi^\star)^{n'} = \int_0^{2\pi} \frac{d\theta}{2\pi} e^{\mathbf{i}(n-n')\theta} \int_0^\infty du e^{-u} u^{\frac{n+n'}{2}}$$

to get $\mathbb{1} = \sum_n |n\rangle\langle n|$.